

Kshitij Agarwal

Data Scientist

Email: kshitij.agarwal.ai@gmail.com | Phone: +1 (672) 962-2482 | Location: Vancouver, BC

LinkedIn: [/kshitij-agarwal](#) | Status: Permanent Resident of Canada

Professional Summary

Data scientist with over **10 years** of experience leveraging an engineering foundation to solve complex challenges through probabilistic and statistical modeling. Proven track record across domains:

- Built a mathematical model of human neurological posture control.
- Designed **real-time closed-loop algorithms** running **24/7** on industrial drilling rigs to detect dysfunctions, **saving millions** of dollars.
- Architected enterprise **Data Quality automation** scanning over **1,000,000** banking datasets.
- Partnered with leading AI labs (OpenAI, Microsoft via Turing) delivering **RLHF** to improve **LLM Python code** generation and **agentic tool-calling** workflows.

Technical Skills

- **Languages:** Python, SQL, R, MATLAB, Java
- **Data & ML / AI:** Machine Learning, Statistical Modeling, RLHF, Agentic AI, PySpark, Pandas, DB2, BigQuery, Hadoop
- **Tools & Platforms:** Google Cloud Platform (GCP), Airflow, Docker, Git, CLI, iOS Development
- **Spoken Languages:** English (Proficient), Hindi (Native)

Relevant Experience

Senior Developer

Turing Enterprises (*Remote*)

Jun 2025 – Dec 2025

- **LLM Code Generation Enhancement (OpenAI / Microsoft):**
 - Addressed code hallucination and syntax errors in early frontier models (**GPT-4** and **o1**).
 - Designed complex coding scenarios and evaluated outputs with rigorous **RLHF feedback**, directly improving the generation of robust, production-grade Python code.
- **Agentic AI Reasoning & Tool Calling (OpenAI / Microsoft):**
 - Targeted multi-step reasoning bottlenecks and **API/tool-calling** failures on complex tasks (e.g., booking workflows, long-context reviews).
 - Supervised and rectified model outputs via RLHF protocols, empowering subsequent model revisions to handle **autonomous multi-step agent** actions.

Manager, Data Quality and Innovations

Bank of Nova Scotia (*Toronto, ON*)

Jun 2022 – Jul 2024

- **Data Quality Automation & Pipeline Architecture:**
 - Modernized legacy manual validation routines across more than **1,000,000** internal banking datasets.
 - Architected **PySpark ETL pipelines** aggregating data from Hadoop, DB2, GCP, and local storage.
 - Deployed **automated Data Profiling** alongside a **Pandas-based anomaly detection** framework incorporating SME feedback loops for remediation.
 - Cut manual processing costs by **\$26 million**, increased data management operational productivity by **40%**, and ensured data integrity for downstream AI products.

Research Engineer

Pason Systems Inc. (*Calgary, AB*)

Aug 2018 – Jun 2022

- **Real-Time Drilling Dysfunction Detection & Mitigation Algorithm:**

- Designed an autonomous, **closed-loop real-time time-series algorithm** in Java to mitigate equipment damage while maximizing drilling speed.
- Achieved a **99% dysfunction detection rate** and drove a **10% increase** in commercial product adoption.
- **Rotary Speed Optimization for Rate-of-Penetration (ROP):**
 - Collaborated with **ExxonMobil** to develop a Java-based algorithm **optimizing drill bit rotary speed (RPM)** in tandem with weight-on-bit (WOB).
 - Validated the system during active field trials across commercial rigs.
- **Predictive ROP Limiting Algorithm:**
 - Built a proactive predictive model in Java that **dynamically capped ROP** prior to dysfunction onset.
- **In-House MATLAB Simulation Platform:**
 - Created a **closed-loop drilling simulator** in MATLAB, reducing dependency on physical QA hardware racks and significantly shortening R&D release cycles.

Research Assistant

University of Alberta (*Edmonton, AB*)

Sep 2016 – Sep 2018

- **Theoretical Model for Human Postural Control:**
 - Investigated disrupted neuromuscular pathways caused by stroke and neurological disorders during seated balance.
 - Executed clinical biomechanical trials, conducted **digital signal processing**, and formulated a theoretical mathematical model explaining neural control of human upright sitting posture.

Other Experience

- **Senior Consultant** | EXL Services (*Gurgaon, HR*) Implemented automated data quality pipelines with proactive anomaly detection to minimize manual oversight.
- **Manager** | Himalaya Developers (*Rishikesh, UK*) Directed end-to-end execution of two residential projects valued at \$3.2M over 18 months (procurement, licensing, civil engineering teams, financial modeling, and sales).
- **iOS Developer** | Pakshi Inc. (*Calgary, AB*) Engineered and shipped two native iOS applications (*More ToLife* and *RoboTask*), managing launch pipelines and digital acquisition strategies.

Education

- **Master of Data Science (MDS)**
University of British Columbia, Vancouver, BC 2026 – Ongoing
- **Master of Science (M.Sc.)**
University of Alberta, Edmonton, AB 2016 – 2018
- **Bachelor of Technology (B.Tech. with Double Major)**
Indian Institute of Technology (IIT) Kanpur, Kanpur, UP 2011 – 2016

Patents & Credentials

Patent

- [US11499411B2](#)

Certifications

- [Advanced Agentic AI](#)
- [LangChain for LLM Development](#)
- [Machine Learning \(Stanford / Coursera\)](#)
- [Machine Learning Engineer Certificate](#)
- [Google Cloud Big Data and Machine Learning Fundamentals](#)
- [Exploring and Preparing your Data with BigQuery](#)